
Endocrine Pharmacology II – Disorders of the Adrenal Gland: Adrenal Insufficiency

 **Carle ILLINOIS**
Carle Illinois College of Medicine

Victoria E. Richards, PhD
verique5@illinois.edu



Learning goal

- **To rationally learn and recall the mechanisms of action, clinical indication, and adverse effects of drugs for the treatment of adrenal insufficiency.**



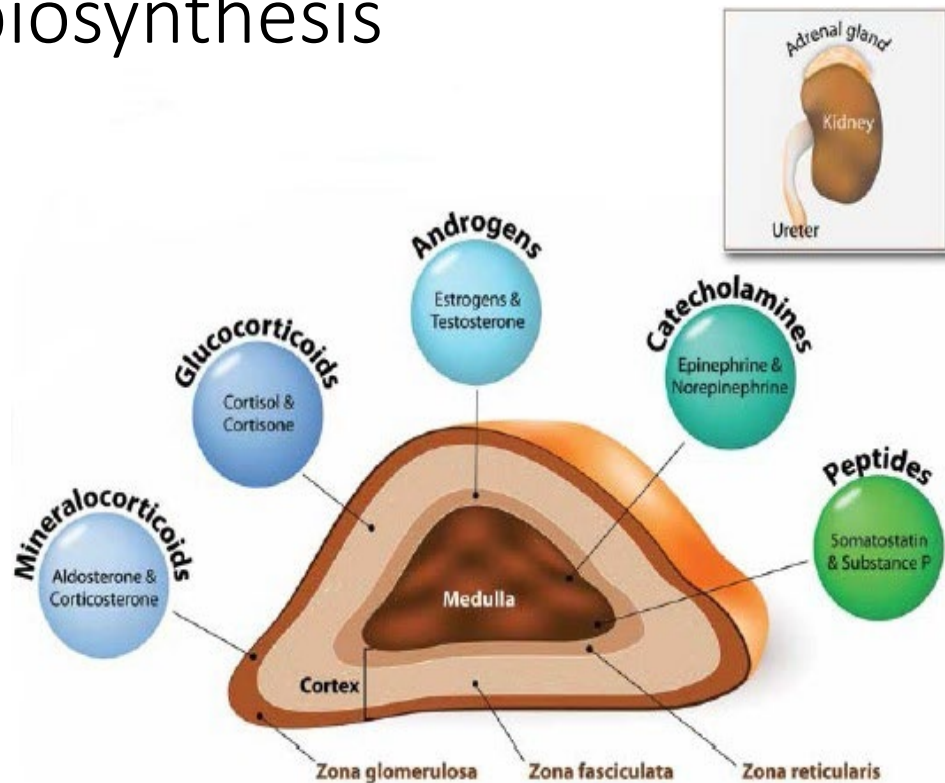
Learning objectives

1. Evaluate the pharmacologic treatment of choice for primary and secondary adrenal insufficiency including the agent of choice and its clinical pharmacology.

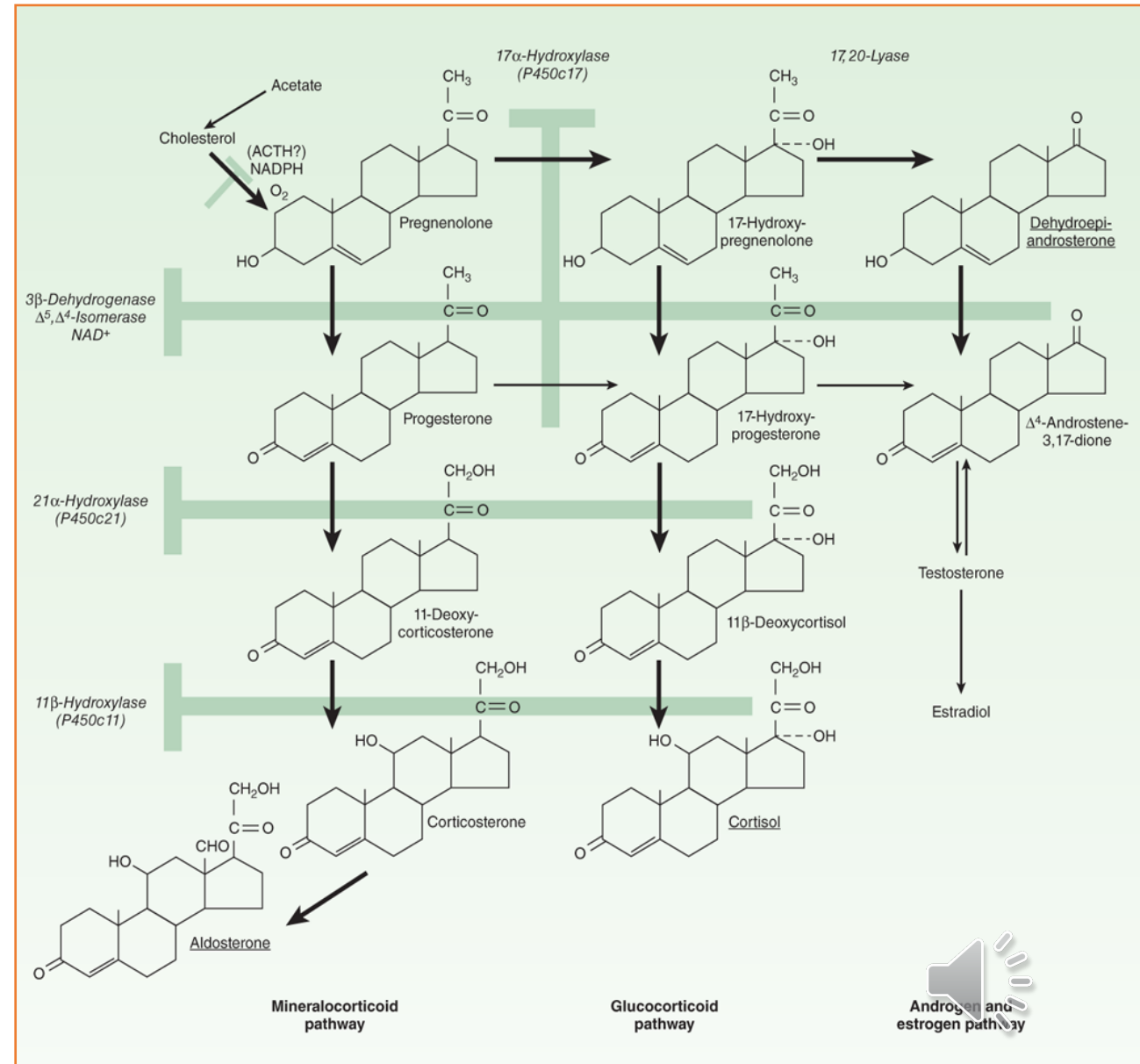


REVIEW

Outline of the major pathways in adrenocortical hormone biosynthesis

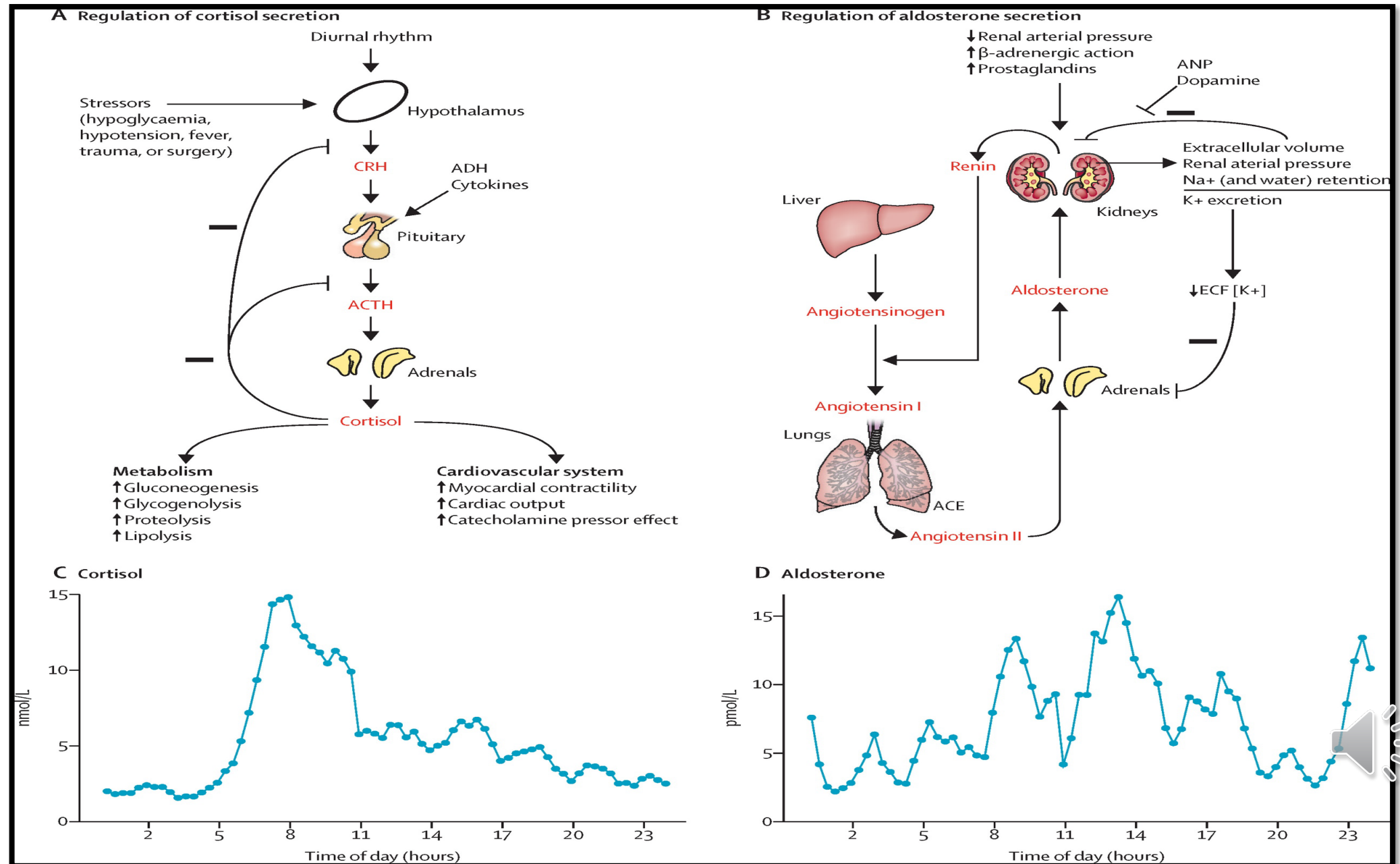


Anderson, 2016



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REVIEW

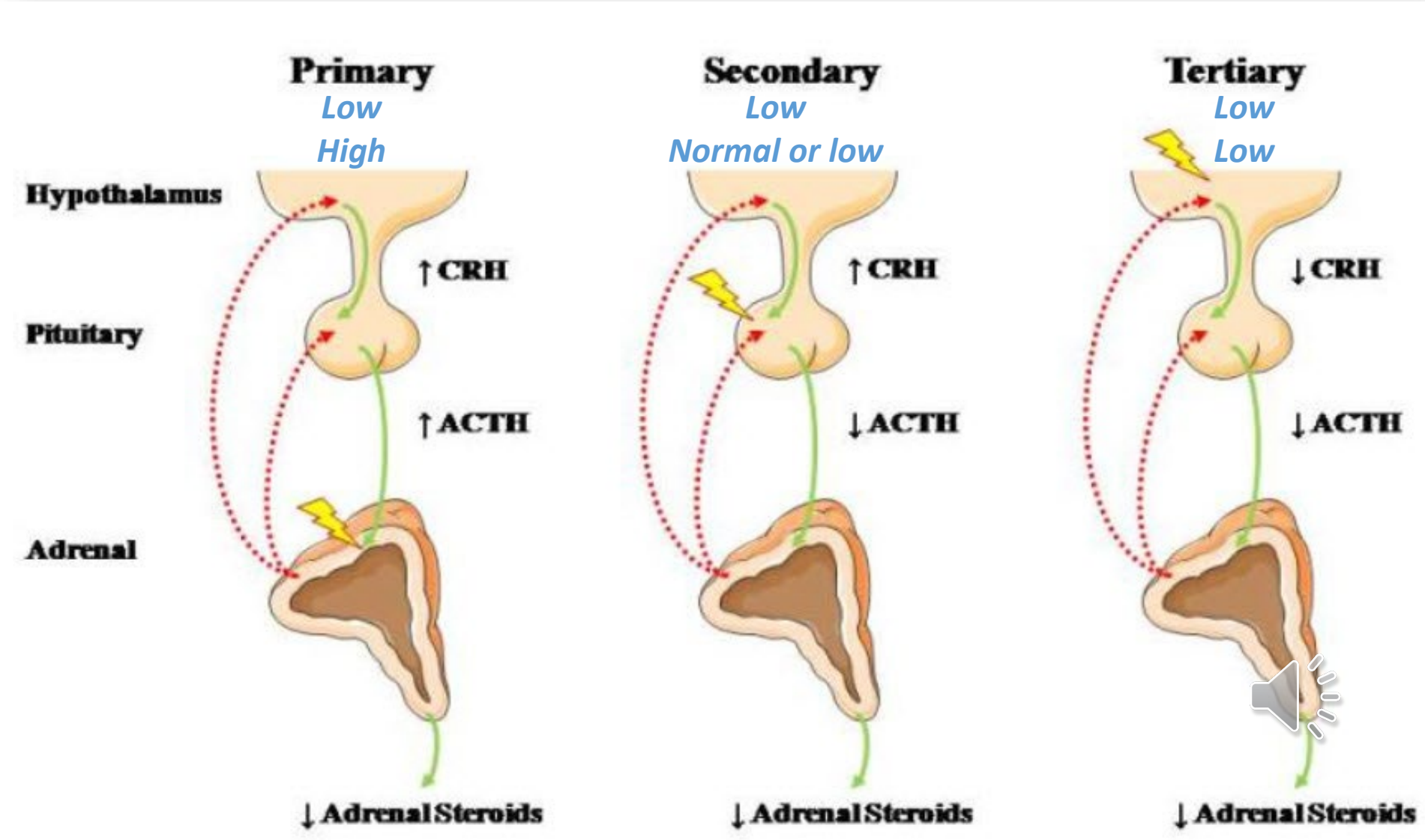


Adrenal insufficiency: Cortisol and Aldosterone

Plasma cortisol production
Plasma [ACTH]

⚡ defect

Endotext



Treatment Goals:

Adrenal Insufficiency

- Replace glucocorticoids and mineralocorticoids in a physiologic manner
- Restore and maintain blood pressure
- Avoid adrenal crises

Mineralocorticoid Deficiency (Hypoaldosteronism)

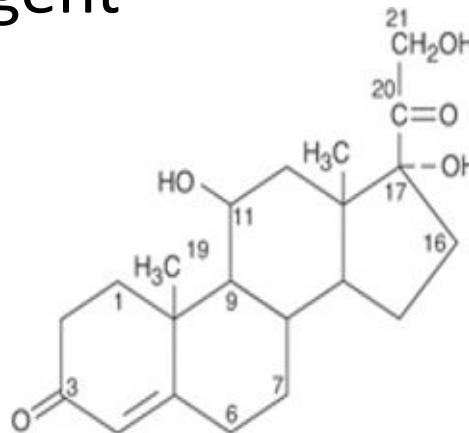
- Due to renin deficiency: sodium replacement or mineralocorticoid



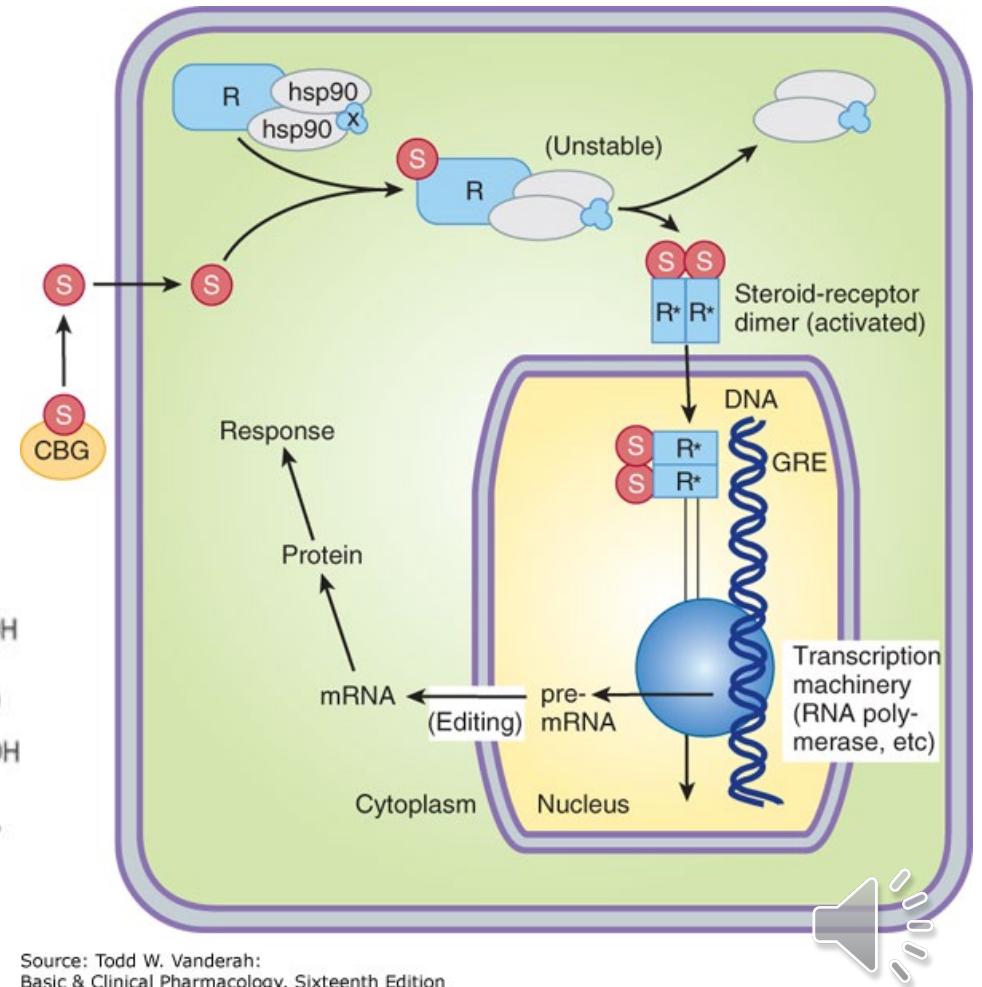
GLUCOCORTICOIDS - REVIEW

Many physiologic effects

- Bind to cytosolic glucocorticoid receptors which translocate to the nucleus
- Circulate in concentrations 100 – 1000X those of aldosterone
- Administration depends on agent
- Metabolized by CYP3As
- Many adverse effects



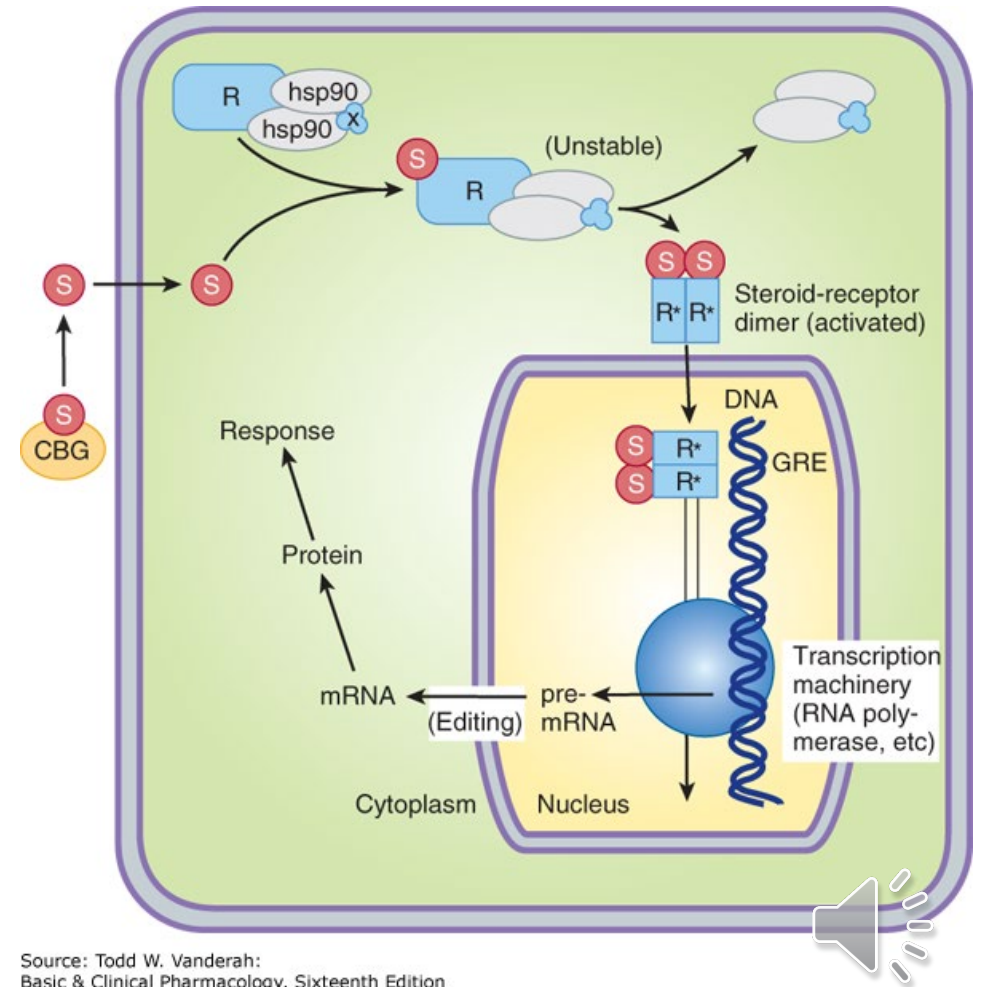
Cortisol (hydrocortisone)



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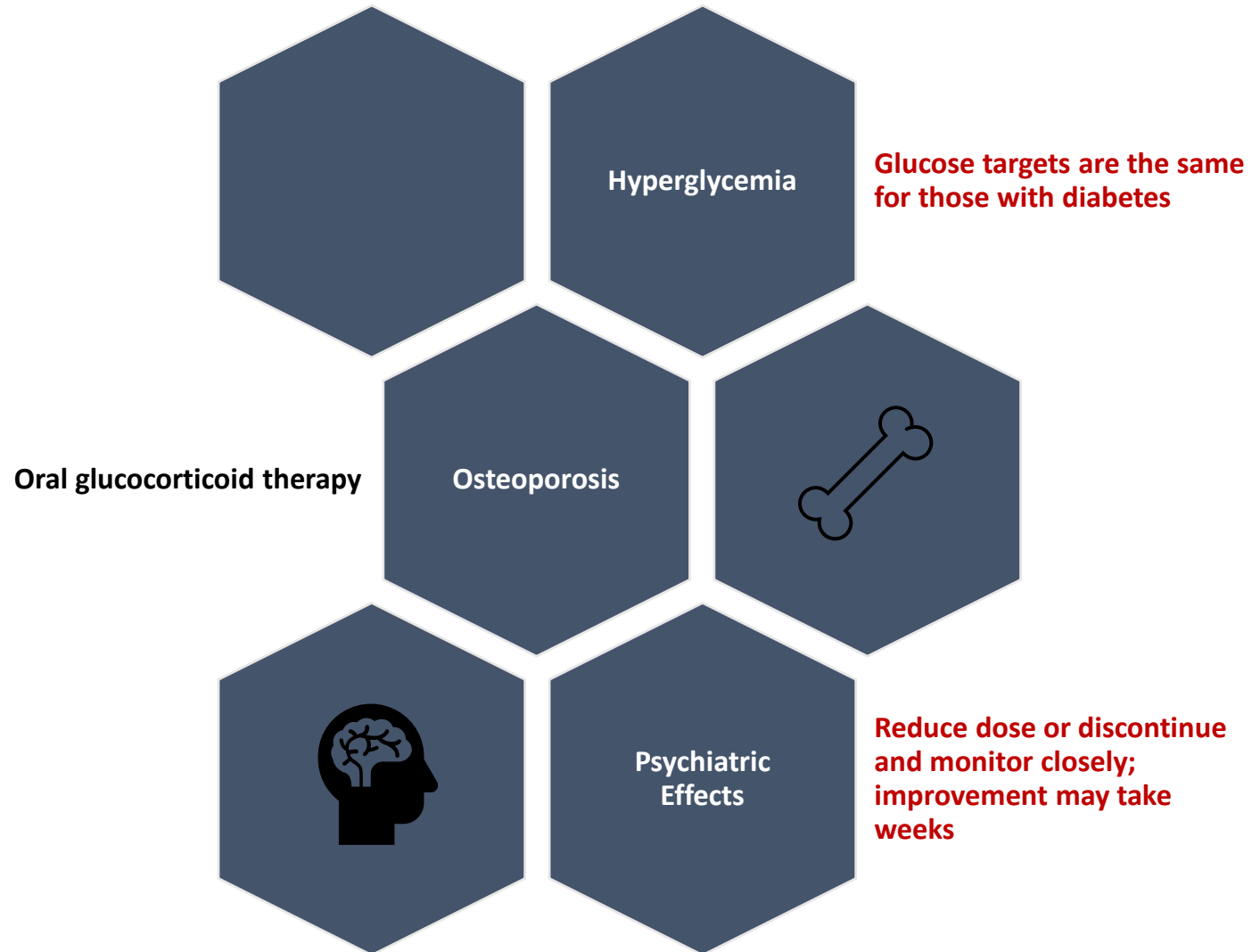
MINERALOCORTICOIDS - REVIEW

- Regulate sodium and water homeostasis
 - Enhance Na^+ and H_2O reabsorption in the kidney, colon and salivary glands
 - Mediated by ENaC, glucocorticoid-regulated kinase (sgk) and the Na^+/K^+ ATPase
- Regulates sodium and potassium transport in the kidney
- MR has similar intrinsic affinity for aldosterone, cortisol, and corticosterone
 - Expressed in many tissues



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Common systemic Corticosteroid Adverse Effects



	Drug	Anti-inflammatory	Salt-retention	Duration
Glucocorticoids as drugs				
	Hydrocortisone	1	2	Short
	Cortisone	0.8	2	Short
	Prednisone	4	1	Mediate (12 – 24 h)
	Prednisolone	4	1	Medium (12 – 24 h)
	Methylprednisolone	5	0	Medium
	Triamcinolone	5	0	Medium
	Dexamethasone	30	0	Long
	Betamethasone	25 - 40	0	Long
Mineralocorticoids as drugs				
	Fludrocortisone	15	125-250	Medium

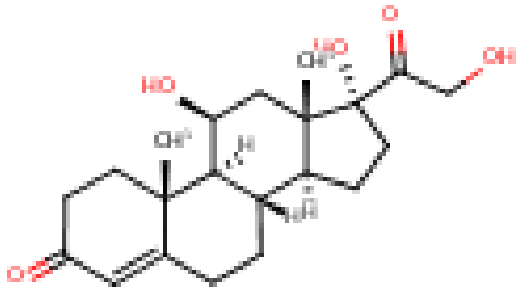


Pharmacologic management of adrenal insufficiency

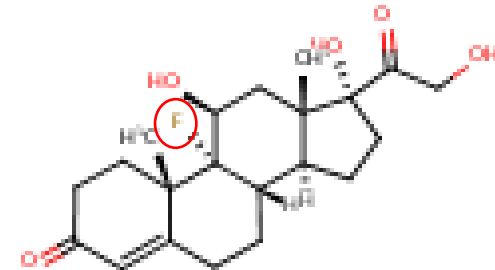
**Preferred
corticosteroid**

Hydrocortisone (or cortisone acetate)*

- Naturally occurring anti-inflammatory
- Antipruritic
- Vasoconstrictive
- Requires multiple daily dosing
- Has salt-retaining properties
- Distributed to skin, muscle, liver, intestines, and kidneys
- Extensively bound to transcortin and albumin
- Does not require hepatic activation
- Can cause osteoporosis



***plus fludrocortisone if mineralocorticoid activity is needed**



Clinical manifestation of Adrenal Insufficiency: **Adrenal crisis**



Presentation is
non-specific

Precipitated by:
Gastroenteritis
Food poisoning
Infection
Surgical and dental
procedures
Injuries
Myocardial infection
Allergic reactions
Severe hypoglycemia in those
with diabetes
Severe psychological stress
Poorly managed AI

- Rapidly life threatening
- Intravenous fluids and **hydrocortisone**



Key points/summary

- Hypoadrenalism is caused by disruption of the hypothalamus, pituitary, or adrenal glands
- Corticosteroids are given in physiologic doses for replacement therapy
 - When given in *supraphysiologic* doses for *prolonged* periods, may cause decreased secretion of endogenous corticosteroids by suppressing pituitary release of corticotropin (2° adrenocortical insufficiency).
 - Drug of choice is hydrocortisone
- Patients with primary adrenal insufficiency typically present with marked impairment of aldosterone secretion
 - Mineralocorticoid replacement is required for full recovery
 - Mineralocorticoid therapy with fludrocortisone is required in 1° but not 2° adrenal insufficiency
 - Aldosterone production is intact in 2°, but *not* 1° adrenal insufficiency
- Patients with secondary adrenal insufficiency should receive replacement cortisol
- During times of stress, patients with known adrenal insufficiency should receive increased doses of glucocorticoid replacement
- Adrenal crisis requires immediate attention. Intravenous hydrocortisone is given, along with fluids, immediately.

